Register No	RA2512044015108

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
21CSC529T – Inferential Statistics
CYCLE TEST – I

Program Offered: M. Tech
Max. Marks: 40

Year / Semester: I / I
Duration: 1.5 Hrs
Date of Exam:

Part - A (2 * 20 = 40 Marks)

Answer any two Questions

Set - B

1. A hospital wants to compare the average daily admissions between two departments (Emergency and General). The admissions data for 5 days from each department is provided below:

Data:

	Day 1	Day 2	Day 3	Day 4	Day 5
Emergency	25	30	22	28	24
General	20	15	18	22	25

Question:

Using the given data:

i. Calculate the measures of central tendency for the given data. (6 marks)
ii. Compute the measures of dispersion for the given data. (6 marks)
iii. Calculate the coefficient of variation for the provided data. (3 marks)

b. Discuss the differences between skewness and kurtosis, and explain their significance. (5 marks)

2.

a. A music festival features three genres: Rock, Pop, and Jazz. The following data shows the number of attendees from different age groups: (5 Marks)

	Rock	Pop	Jazz
Adult	70	50	30
Teen	60	80	40
Child	50	30	60

i. What is the probability that a randomly chosen attendee is enrolled in Pop?
ii. What is the probability that a randomly chosen attendee is a Teen?
iii. What is the probability that a randomly chosen attendee is an Adult who prefers Rock?
iv. What is the probability that a randomly chosen attendee is a Child who enjoys Jazz?
v. What is the probability that an attendee is an Adult given that the attendee prefers Rock?

b. A clothing manufacturer produces three types of garments - shirts, pants, and jackets. The probability of producing a defective shirt is 0.02, a defective pair of pants is 0.03, and a defective jacket is 0.04. The probabilities that a defective item is a shirt, a pair of pants, or a jacket are 0.5, 0.3, and 0.2, respectively. If a defective item is selected from the production line, what is the most likely type of garment? (10 Marks)

c. Outline the different types of non probabilistic sampling techniques and also provide examples for each (5 Marks)

3.

a. A fitness researcher is investigating the effects of two factors, 'Diet Quality' and 'Exercise Frequency,' on individuals' 'Fitness Level.' The following data is collected for 5 individuals over a specific time period:

S.NO	Diet Quality (scale of 1-10)	Exercise Frequency (days per week)	Fitness Level Score (out of 100)
1	8	4	85
2	5	2	60
3	9	5	90
4	6	3	70
5	4	1	50

Determine which independent variable (Diet Quality or Exercise Frequency) has a stronger relationship with Fitness Level. (10 marks)

b. Consider the dataset: [12, 15, 22, 25, 29, 34, 38]. Calculate the quartile Q1, Q2, Q3, and the IQR (Interquartile Range). (5 Marks)

c. A local choir has 12 members, and they want to form a committee of 5 members to organize an upcoming concert. How many different ways can the choir members form this committee? (5 Marks)

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